

HybridStack: Explainable hybrid ensemble for accurate inflation forecasting with high-frequency macro-financial data

S. M. Mohiuddin Khan Shiam ^{a,e}, Meherunnesa Neela ^{b,e}, Amrijit Biswas ^{c,e},
Mahdy Rahman Chowdhury ^{c,d},*

^a Department of Computer Science and Engineering, BRAC University, Dhaka, 1212, Bangladesh

^b Department of Mathematics, Jahangirnagar University, Savar, Dhaka, 1342, Bangladesh

^c Department of Electrical and Computer Engineering, North South University, Bashundhara R/A, Dhaka, 1229, Bangladesh

^d NSU Center of Quantum Computing, North South University, Bashundhara R/A, Dhaka, 1229, Bangladesh

^e Mahdy Research Academy, Dhaka, Bangladesh

ARTICLE INFO

Dataset link: <https://github.com/mohiuddin-kh-an-shiam/HybridStack>

Keywords:

Inflation forecasting
Macroeconomic indicators
Machine learning
Deep learning
Explainable AI

ABSTRACT

Inflation forecasting plays a critical role in macroeconomic policy and financial decision-making. Accurate forecasts help stabilize economies amid uncertainty. Traditional models often struggle to capture nonlinear dynamics, high-dimensional data, and sudden regime shifts, leading to ineffective policy decisions. This study addresses these gaps by proposing HybridStack, a novel hybrid stacking ensemble combining gradient-boosted trees with regularized linear regression, optimized through Bayesian hyperparameter tuning and meta-learning strategies. The model is trained on a harmonized high-frequency panel dataset constructed by merging ten macro-financial sources covering U.S. inflation from 2006 to 2024. Empirical results show that HybridStack outperforms fifteen baseline models across seven metrics. On the held-out test set, it achieves a root mean squared error (RMSE) of 0.0006, a mean absolute error (MAE) of 0.0004, and a coefficient of determination (R^2) of 0.9999. Notably, HybridStack achieves approximately a 99.92% reduction in RMSE relative to the best-performing prior U.S. study benchmark, a 94.7% reduction compared to the standalone Ridge baseline, and a 97.6% reduction relative to XGBoost on the held-out test set, establishing a new state-of-the-art benchmark within our dataset and evaluation framework. Further explainable AI (XAI) analyses using SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), and Partial Dependence Plots (PDP) highlight the primacy of short-term lags and rolling means as dominant predictors, with inflation expectations exhibiting episodic yet influential effects. This work contributes a transparent, empirically validated forecasting tool for policymakers and researchers, enhancing the reliability and interpretability of inflation predictions in volatile economic environments. The data and code used in this study are available at <https://github.com/mohiuddin-khan-shiam/HybridStack>.

1. Introduction

Headline inflation in advanced economies surged to 9.5% in the second quarter of 2022, its highest level in four decades, before sliding to 2.3% by the end of 2023 [1]. Inflation forecasting constitutes a cornerstone of macroeconomic policy and financial decision-making, influencing central bank strategies, investment planning, and fiscal adjustments worldwide. In an era marked by unprecedented economic volatility, such as the disruptions from the 2008 Global Financial Crisis and the 2020–2022 COVID-19 pandemic, accurate predictions are essential for mitigating risks and stabilizing economies. Traditional econometric models [2,3], including autoregressive [4] and vector

autoregressive [5] approaches, have long dominated this field due to their theoretical coherence and transparency. However, these linear methods often falter in the presence of nonlinear dynamics, high-dimensional data, or abrupt regime shifts. This is highlighted in the European Central Bank's analysis of the 2021–2022 inflationary surge [6]. While autoregressive and vector autoregressive models serve as foundational references in the literature, they are not employed as empirical baselines in this study. Recent advancements in machine learning have introduced promising alternatives, leveraging algorithms like random forests, gradient boosting, and deep neural networks to capture complex patterns in macroeconomic indicators. These methods have

* Corresponding author.

E-mail addresses: mohiuddin.khan.shiam@g.bracu.ac.bd (S.M.M.K. Shiam), 20210152524neela@juniv.edu (M. Neela), amrijit.biswas01@northsouth.edu (A. Biswas), mahdy.chowdhury@northsouth.edu (M.R. Chowdhury).

<https://doi.org/10.1016/j.array.2026.101003>

Received 21 August 2025; Received in revised form 4 June 2026; Accepted 9 June 2026

Available online 19 June 2026

2590-0056/© 2026 Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

demonstrated superior performance in diverse contexts, ranging from emerging markets in Asia to advanced economies such as the United States. This is evidenced by studies incorporating high-frequency data and hybrid frameworks to enhance predictive accuracy.

Despite these progresses, significant gaps persist in the literature, particularly regarding the integration of explainability and robustness across economic regimes. This trade-off is particularly consequential in finance. Černevičienė and Kabainkas [7] systematically reviewed 138 XAI studies and confirmed that opacity in financial AI directly undermines trust, regulatory compliance, and accountability. Arsenault et al. [8] provide a recent survey of explainable AI methods for financial time-series forecasting, highlighting how inadequate explanations impede adoption in high-stakes environments. Furthermore, while some investigations address structural breaks or high-frequency inputs, they often overlook comprehensive harmonization of asynchronous data or systematic benchmarking against a wide array of baselines. This limitation is compounded by a predominant focus on low-frequency aggregates, which mask intra-period fluctuations and asymmetric transmissions from variables like inflation expectations and commodity prices. Consequently, there remains a critical need for frameworks that not only achieve state-of-the-art accuracy but also provide transparent insights into underlying drivers, thereby bridging the divide between algorithmic sophistication and practical utility in volatile economic environments.

This study addresses these challenges by proposing a novel hybrid stacking ensemble, termed HybridStack. The model synergizes gradient-boosted trees and regularized linear regression to forecast U.S. inflation using a harmonized high-frequency panel of ten macro-financial indicators spanning 2006 to 2024. By employing Bayesian optimization and meta-learning, our approach captures both linear and nonlinear dependencies while ensuring computational efficiency. Empirical evaluations reveal that HybridStack outperforms 15 baselines across seven metrics, with explainable AI techniques highlighting the primacy of short-term lags and rolling means. The remainder of this paper is structured as follows: Section 2 reviews the literature, Section 3 details the methodology, Section 4 describes the experiments, Section 5 presents the results, Section 6 discusses interpretations and insights, and Sections 7 and 8 conclude with limitations and future directions.

Our main contributions are listed below:

- **Novel macro-financial panel:** We assemble a high-frequency U.S. panel of ten indicators, such as inflation expectations, prices, real activity, labor-market slack, policy/term rates, exchange rates, and energy, spanning 2006–2024, enabling reproducible, regime-spanning analysis.
- **Bespoke hybrid architectures:** We propose a hybrid model, HybridStack, which integrates gradient-boosted trees with regularized linear regression. This model is designed to simultaneously capture nonlinear and linear dependency structures within high-dimensional feature spaces. By leveraging the complementary strengths of both model families, HybridStack enhances robustness while improving predictive accuracy across heterogeneous and dynamic data environments.
- **Robustness across regime shifts:** We address the sudden shifts in inflation that characterized the 2008–2009 financial crisis and the 2020–2022 pandemic. To facilitate a robust assessment of stability across these distinct economic regimes, we employ rolling/walk-forward validation spans.
- **Policy-relevant empirical insights:** We establish a state-of-the-art result in inflation forecasting by benchmarking our HybridStack model against 15 established baselines, achieving superior performance across all evaluated metrics. Explainability analyses consistently prioritize short-horizon inflation lags and rolling means as dominant features, while inflation expectations contribute episodic signals in the model's predictive framework.

Table 1

The table shows the geographical context (country or countries involved) and the time span of the datasets used for each existing study.

Authors	Country	Dataset period
Liu Y. et al. [9]	Japan	2012–2023
Kanaparthi V. et al. [10]	United States	1968–2022
Naghi A. A. et al. [11]	US, Canada, UK	2016–2022
Narayanaa T. L. et al. [12]	United States	1947–2021
Ayuputri A. A. et al. [13]	Brunei, Malaysia, Thailand, Indonesia, Vietnam	1960–2022
Li S. et al. [14]	United States	1980–2023
Arif E. et al. [15]	Indonesia	2009–2022
Xavier A. L. S. et al. [16]	Brazil	1999–2023
Jyothi Rani S. A. et al. [17]	India	2009–2021
Akbulut H. et al. [18]	Turkey	2006–2020
Nakorji M. et al. [19]	Nigeria	2021
Das P. K. et al. [20]	India	2012–2022
Ismail L. B. et al. [21]	Bangladesh	2013–2022
Amirthavarshini V. et al. [22]	India	1993–2023
Kalaiselvi K. et al. [23]	India	2013–2023

2. Literature review

The empirical and methodological landscape of inflation forecasting has evolved rapidly in recent years, driven by the growing availability of high-frequency data and the versatility of machine-learning algorithms. A critical examination of contemporary studies reveals three interrelated strands of scholarship: (i) single-country analyses in emerging and advanced economies, (ii) comparative assessments that juxtapose traditional econometric baselines with modern machine-learning or deep-learning models, and (iii) hybrid frameworks that integrate linear and nonlinear representations to accommodate regime shifts and structural breaks. Table 1 outlines the geographical scope and temporal coverage of the datasets employed across the reviewed works, thereby contextualizing the diversity of economic environments under study.

2.1. Machine learning in emerging-economy contexts

Several investigations concentrate on inflation dynamics in South and Southeast Asia, where structural heterogeneity and data limitations present acute modeling challenges. Kalaiselvi et al. examine India's post-COVID inflation trajectory using a spectrum of algorithms, ranging from linear regression and k -nearest neighbors to Random Forests (RF) and Long Short-Term Memory networks (LSTM) [23]. Their findings underscore the predictive superiority of RF, which yielded an exceptionally high coefficient of determination, while also exposing the constraints imposed by a relatively narrow feature set. Extending the Indian context, Amirthavarshini et al. demonstrate that a Bidirectional LSTM (BiLSTM) trained on three decades of Consumer Price Index data attains the lowest mean squared error, thereby highlighting the sensitivity of deep recurrent architectures to long memory and seasonality [22]. Das et al. further corroborate the merits of nonlinear tree ensembles by showing that RF outperforms autoregressive integrated moving-average models when confronted with a high-dimensional predictor matrix that encompasses fifty-six macro-financial indicators [20].

Neighboring Bangladesh exhibits analogous benefits from deep learning. Ismail et al. benchmark LSTM and XGBoost against classical statistical methods, reporting improvements across root mean squared error, mean absolute percentage error, and mean absolute error [21]. In West Africa, Nakorji et al. find that ridge regression and feed-forward neural networks provide accurate forecasts of Nigerian inflation once food, core, and lending-rate determinants are incorporated, though the authors caution that monetary policy inefficiencies can attenuate model reliability [19]. Collectively, these studies affirm the adaptability

Table 2

The table lists the existing machine-learning or statistical models employed in each inflation-forecasting study.

Authors	Models used
Liu Y. et al. [9]	LASSO, Elastic Net, RF, XGBoost
Kanaparthi V. et al. [10]	RF, AdaBoost, Regularized Regressions
Naghi A. A. et al. [11]	RF, GBT, BART, SVM, NN, Penalized Linear Regressions
Narayanaa T. L. et al. [12]	ARIMA, LSTM
Ayuputri A. A. et al. [13]	NN Backpropagation, SVM, RNN-LSTM
Li S. et al. [14]	LSTM, SARIMA, DWT
Arif E. et al. [15]	ARFIMA, LSTM
Xavier A. L. S. et al. [16]	ARIMA, ETS, MLP, SVR, RF, ETS+Bagging
Jyothi Rani S. A. et al. [17]	ARIMA, MLP, RNN, CNN, LSTM
Akbulut H. et al. [18]	VAR, Ridge Regression, LASSO, MLP
Nakorji M. et al. [19]	Ridge Regression, LASSO, Elastic Net, PLS, ANN
Das P. K. et al. [20]	Ridge Regression, LASSO, Elastic Net, RF, ANN
Ismail L. B. et al. [21]	ARIMA, LSTM, GRU, SVR, XGBoost (AutoML), Prophet, SARIMA
Amirthavarshini V. et al. [22]	LSTM, BiLSTM, XGBoost, Random Forest
Kalaiselvi K. et al. [23]	Linear Regression, KNN, RF Regression, GBDT, LSTM

of machine-learning models to heterogeneous economic environments, yet they concur on two persistent limitations: inadequate coverage of external shocks and the prospective risk of overfitting in data-scarce settings.

2.2. Comparative evidence from volatile and developed economies

High-volatility economies offer a fertile ground for evaluating the resilience of forecasting techniques. Akbulut et al. analyze Turkish inflation between 2006 and 2020 and observe that vector autoregression retains an edge over penalized linear regressions and multilayer perceptrons in root mean squared error terms, though the nonlinear multilayer perceptron narrows the performance gap [18]. In Brazil, Xavier et al. advocate a hybrid exponential-smoothing state-space model enhanced by bagging; their ensemble surpasses individual models and alternative hybrids on multiple error metrics, illustrating how variance-reduction strategies can stabilize forecasts in turbulent markets [16].

Advanced economies reveal complementary insights. Kanaparthi et al. evaluate United States inflation during the COVID-19 period and conclude that RF and AdaBoost outperform linear benchmarks, especially in high-inflation sub-samples, while Shapley values reaffirm the enduring relevance of a Phillips-curve relationship [10]. Focusing on regime heterogeneity, Liu et al. demonstrate that LASSO, rather than nonlinear ensembles, delivers the lowest error for Japanese core inflation across multiple horizons, thereby emphasizing the prudence of parsimonious penalty functions in low-variance settings [9]. Narayanaa et al. add a temporal dimension by showing that autoregressive integrated moving-average models dominate short-run United States forecasts, whereas LSTM excels over medium horizons, implying that model efficacy is contingent upon forecast length and data granularity [12]. Table 2 provides a concise inventory of the modeling techniques employed across studies.

2.3. Hybrid and decomposition frameworks

Hybrid frameworks seek to reconcile linear interpretability with nonlinear flexibility. Arif et al. couple an autoregressive fractionally integrated moving-average backbone with LSTM residual modeling to capture Indonesia's long memory and volatility; the composite architecture consistently surpasses both ARFIMA and ARFIMA-GARCH comparators across multiple error metrics [15]. Li et al. introduce a discrete wavelet transform pre-processing layer prior to an LSTM-seasonal autoregressive integrated moving-average ensemble, revealing that the multi-resolution decomposition significantly enhances United States forecast accuracy during inflationary spikes [14]. Extending hybrid integration directly to U.S. CPI forecasting, Gur [24] demonstrates that an LSTM-XGBoost architecture, where LSTM extracts temporal feature representations subsequently refined by gradient-boosted trees, optimized

through Bayesian and genetic algorithm methods, consistently outperforms standalone LSTM, MARS, and XGBoost models across RMSE, MAE, MAPE, and R^2 metrics, with SHAP analysis confirming that the federal funds rate and GDP are the dominant predictive drivers of consumer price dynamics. Naghi et al. provide multicountry evidence by testing more than thirty algorithms for the United States, Canada, and the United Kingdom; their results confirm Random Forest supremacy under normal conditions but identify boosted trees and stochastic-volatility models as more robust during the 2020–2022 shock [11]. Finally, Ayuputri et al. employ neural networks, support vector machines, and recurrent LSTM architectures to forecast post-pandemic inflation in five ASEAN economies, establishing that well-tuned backpropagation networks achieve exceptional accuracy, albeit at the expense of potential overfitting in the absence of exogenous covariates [13].

2.4. Synthesis and research gaps

The reviewed studies converge on three salient observations. First, nonlinear machine-learning and deep-learning algorithms frequently surpass classical econometric models, particularly when structural breaks or high-dimensional predictors are present. Second, model efficacy is sensitive to country-specific inflation regimes and data horizons, necessitating flexible frameworks that adjust to evolving macroeconomic conditions. Third, despite notable accuracy gains, many investigations acknowledge deficiencies in interpretability, external shock accommodation, and hybrid integration. The forthcoming empirical analysis addresses these gaps by benchmarking a diverse portfolio of algorithms, emphasizing explainable artificial-intelligence techniques, and adopting a two-stage hybrid architecture that balances transparency with predictive power.

3. Methodology

3.1. Challenges in inflation forecasting

The main challenges in inflation forecasting are as follows:

- 1. Limitations of Linear Frameworks:** Models fail to capture nonlinear dynamics beyond linear assumptions, leading to persistent forecasting errors.
- 2. Isolated Analysis and Frequency Mismatches:** Indicators are examined in isolation or low-frequency panels, missing asymmetric cross-frequency transmissions from inflation expectations and commodity volatility.
- 3. Obscuration by Data Aggregation:** Aggregated data masks high-frequency fluctuations (e.g., daily changes in breakeven rates, oil prices, and labor slack), causing nonlinear amplifications and reversals.

4. **Inadequate Integration of High-Frequency Data:** High-frequency variables are rarely aligned with macroeconomic series, filtering out intra-month signals and underestimating inflation dynamics.
5. **Unidentified Latent Thresholds:** Latent thresholds from joint indicators remain undetected, limiting forecast accuracy and policy credibility.

3.2. Proposed approach

To address the identified challenges, we introduce a novel framework that integrates high-frequency data harmonization with an advanced hybrid modeling technique. This approach enhances inflation forecasting by capturing nonlinear dynamics and cross-frequency interactions.

Harmonized High-Frequency Dataset: The proposed framework harmonizes ten high-frequency macroeconomic indicators, including market-based inflation expectations, real-sector activity proxies, commodity prices, and interest-rate term structures, aligned on a daily grid spanning 2006 to 2024. Asynchronous data gaps are mitigated through time-restricted forward filling and linear interpolation, with all series standardized and augmented by lagged terms, and rolling means. By merging these diverse datasets, the framework provides a comprehensive view of inflation dynamics, enabling more accurate representation of real-time economic signals.

Proposed Hybrid Forecasting Model: Hybrid models have demonstrated superior performance over baseline approaches in capturing complex patterns. In this study, we develop and evaluate a novel hybrid model designed to outperform traditional benchmarks by integrating machine learning and econometric techniques. This model is computationally efficient, requiring low resources and minimal training time, making it accessible for widespread adoption in academic and policy applications.

3.3. Base models

Fig. 1 illustrates the overall workflow for inflation forecasting, encompassing data collection, preprocessing, model training, evaluation, and interpretation. This section provides detailed explanations of the forecasting models, which are organized according to the broad categories shown in the workflow: distance-based, kernel-based, linear and regularized, tree-based, and deep learning. Each approach is described with attention to its theoretical foundation and relevance for capturing macroeconomic signals that influence inflation.

Distance-Based Model:

K-Nearest Neighbors Regression (KNN): KNN predicts $\hat{y}(\mathbf{x})$ by averaging the K nearest neighbors' responses:

$$\hat{y}(\mathbf{x}) = \frac{1}{K} \sum_{\mathbf{x}_i \in \mathcal{N}_K(\mathbf{x})} y_i. \quad (1)$$

It captures local patterns in macroeconomic data but scales poorly with high dimensions.

Kernel-Based Model:

Support Vector Regression (SVR): SVR minimizes $\frac{1}{2} \|\mathbf{w}\|^2 + C \sum (\xi_i + \xi_i^*)$ subject to deviation constraints within ϵ . Kernels enable nonlinear mappings, revealing inflation patterns from economic indicators.

Linear and Regularized Models:

ElasticNet Regression (EN): ElasticNet balances L1 and L2 penalties:

$$\min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2. \quad (2)$$

It selects and shrinks features in correlated macroeconomic sets.

Lasso Regression (Lasso): Lasso enforces sparsity via L1 penalty:

$$\min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_1. \quad (3)$$

It identifies key inflation drivers.

Ridge Regression (Ridge): Ridge shrinks via L2 penalty:

$$\min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_2^2. \quad (4)$$

It addresses multicollinearity in macro variables.

Tree-Based Models:

Single Decision Tree Regressor: Predicts via mean in partitioned regions:

$$f_{\text{DTR}}(\mathbf{x}) = \sum_{k=1}^K c_k \mathbb{I}(\mathbf{x} \in R_k). \quad (5)$$

Transparent but prone to overfitting interdependent variables.

Bagging Methods: Bagging averages trees from bootstrapped samples to reduce variance.

Extremely Randomized Trees Regressor (ERT): Adds randomness in splits for diversity.

Random Forest (RF): Averages bootstrapped trees with feature subsampling:

$$\hat{y}_{\text{RF}}(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M h_m(\mathbf{x}). \quad (6)$$

Captures economic interactions effectively.

Boosting Methods: Boosting sequentially fits trees to residuals.

Gradient Boosting Regressor (GBR): Updates as $F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta h_m(\mathbf{x})$, adapting to economic shifts.

Categorical Boosting (CatBoost): Handles categories and numerics with ordering to limit overfitting in volatile data.

Extreme Gradient Boosting (XGBoost): Adds regularization and optimized structures to control complexity.

Histogram-Based Gradient Boosting Regressor (HGBR): Uses histograms for efficient splits in large macro tasks.

Light Gradient Boosting Machine (LightGBM): Employs leaf-wise growth, GOSS, and EFB for high-dimensional efficiency.

Deep Learning Models:

Bidirectional Gated Recurrent Unit (Bi-GRU): Bi-GRU uses update and reset gates for sequential data, processing bidirectionally:

$$\mathbf{z}_t = \sigma(\mathbf{W}_z \mathbf{x}_t + \mathbf{U}_z \mathbf{h}_{t-1}), \quad (7)$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r \mathbf{x}_t + \mathbf{U}_r \mathbf{h}_{t-1}), \quad (8)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h \mathbf{x}_t + \mathbf{U}_h (\mathbf{r}_t \odot \mathbf{h}_{t-1})), \quad (9)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t. \quad (10)$$

It detects leading and lagging macroeconomic effects effectively.

Bidirectional Long Short-Term Memory (BiLSTM): BiLSTM maintains cell states for long-term dependencies, combining forward and backward states:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f), \quad (11)$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i), \quad (12)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c \mathbf{x}_t + \mathbf{U}_c \mathbf{h}_{t-1} + \mathbf{b}_c), \quad (13)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \quad (14)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{b}_o), \quad (15)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t). \quad (16)$$

It models varied timescales in inflation, integrating trends and shocks.

In summary, these models, from distance- and kernel-based to linear, tree ensembles, and deep networks, address inflation forecasting challenges.

3.4. Proposed hybrid architecture

We propose a principled hybrid ensemble architecture that integrates heterogeneous base learners, gradient-boosted decision trees (XGBoost) and Ridge regression, through a meta-learning framework.

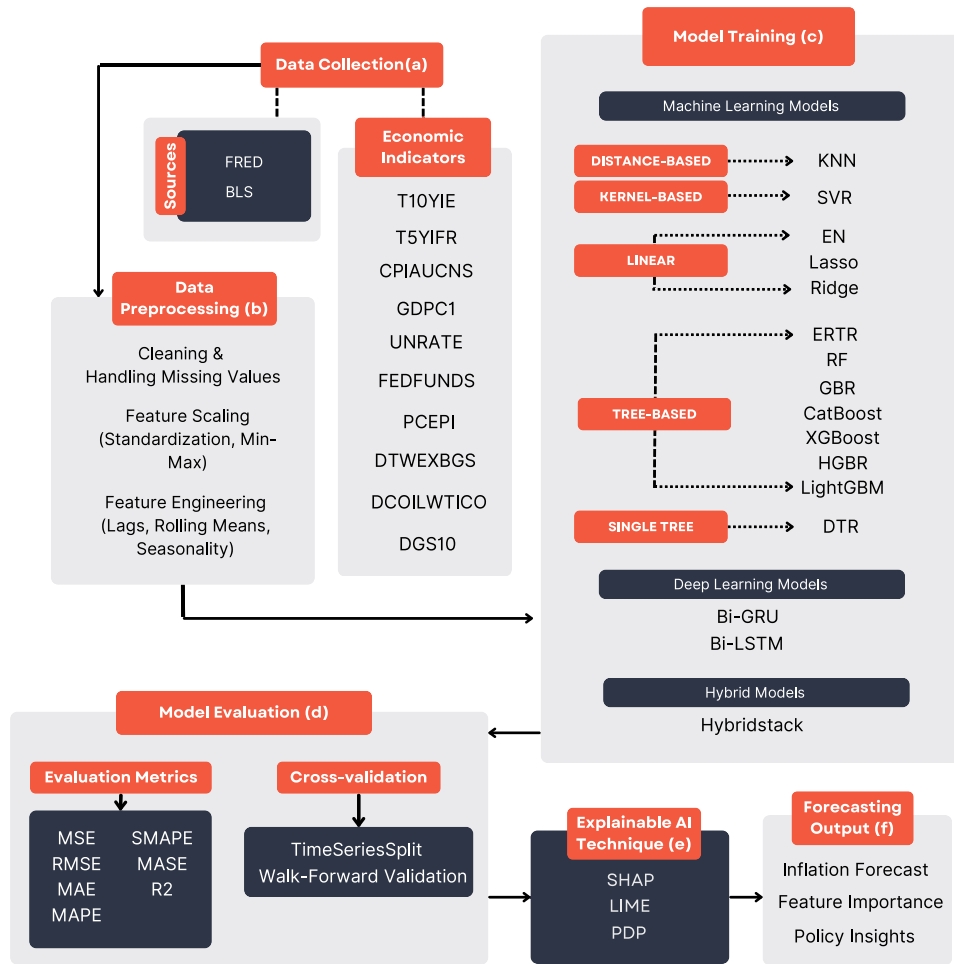


Fig. 1. Schematic overview of our proposed forecasting framework: (a) Data collection is initiated using Federal Reserve Economic Data (FRED) and Bureau of Labor Statistics (BLS) repositories, capturing a range of relevant economic indicators; (b) preprocessing integrates rigorous cleaning and advanced feature engineering to enhance temporal signal extraction; (c) a diverse suite of classical, ensemble, and deep learning models is employed for robust training; (d) performance assessment utilizes multiple validation strategies and error metrics; (e) explainable AI techniques are incorporated to enhance interpretability and transparency; and (f) the final outputs deliver actionable multi-step forecasts, feature attribution, and policy-relevant insights.

The architecture, depicted in Fig. 2, is designed to capture both non-linear and linear dependency structures in high-dimensional feature spaces, improving robustness and predictive accuracy across varying data landscapes.

Model Structure and Theoretical Foundation:

Let $D = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ denote the set of training examples, where $\mathbf{x}_i \in \mathbb{R}^p$ is the input feature vector and $y_i \in \mathbb{R}$ is the target response. The modeling pipeline is organized as follows:

Base Learners:

The proposed ensemble includes two distinct base models, each optimized using Bayesian strategies to achieve complementary modeling strength.

XGBoost Regressor (XGB): The first base learner, $\mathcal{M}_{\text{XGB}}$, consists of an additive ensemble of regression trees trained via gradient boosting. Each prediction is computed as

$$f_{\text{XGB}}(\mathbf{x}) = \sum_{m=1}^M f_m(\mathbf{x}; \theta_m), \quad (17)$$

where f_m represents the m th regression tree, parameterized by θ_m . Tree complexity and regularization parameters are automatically tuned using Bayesian optimization.

Ridge Regression (Ridge): The second base learner, $\mathcal{M}_{\text{Ridge}}$, is a linear regression model with L_2 regularization. Ridge regression estimates the

coefficients β by solving

$$\beta_R^* = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \alpha \|\beta\|_2^2, \quad (18)$$

where $\mathbf{X}$ is the feature matrix, α is the regularization parameter, and $\|\cdot\|_2$ denotes the Euclidean norm.

Bayesian Hyperparameter Optimization:

Hyperparameters for both learners are selected using Bayesian optimization. Unlike exhaustive grid search, Bayesian optimization builds a probabilistic surrogate model of the objective function and iteratively selects the next hyperparameter configuration most likely to improve performance, thereby drastically reducing the number of costly cross-validation evaluations required. For a model $\mathcal{M}_k$ parameterized by λ_k from the search space Λ_k , the optimal parameters are computed as follows:

$$\lambda_k^* = \arg \min_{\lambda \in \Lambda_k} \mathbb{E}_{\mathcal{F}} [\mathcal{L}_{\text{CV}}(\mathcal{M}_k(\cdot; \lambda), \mathcal{D}_{\text{train}})], \quad (19)$$

where $\mathcal{L}_{\text{CV}}$ denotes the average cross-validated loss over the folds in $\mathcal{F}$.

In practice, this optimization was implemented via Optuna and `gpmimimize`, with pruning of unpromising trials to further reduce computational overhead 4.4.

Stacked Generalization Using Meta-Learning:

Following hyperparameter selection and training, the base models produce out-of-fold (OOF) predictions for each training example. Formally, for $i = 1, \dots, N$, let

$$z_{\text{XGB},i} = f_{\text{XGB}}^{(-k(i))}(\mathbf{x}_i), \quad z_{\text{Ridge},i} = f_{\text{Ridge}}^{(-k(i))}(\mathbf{x}_i),$$

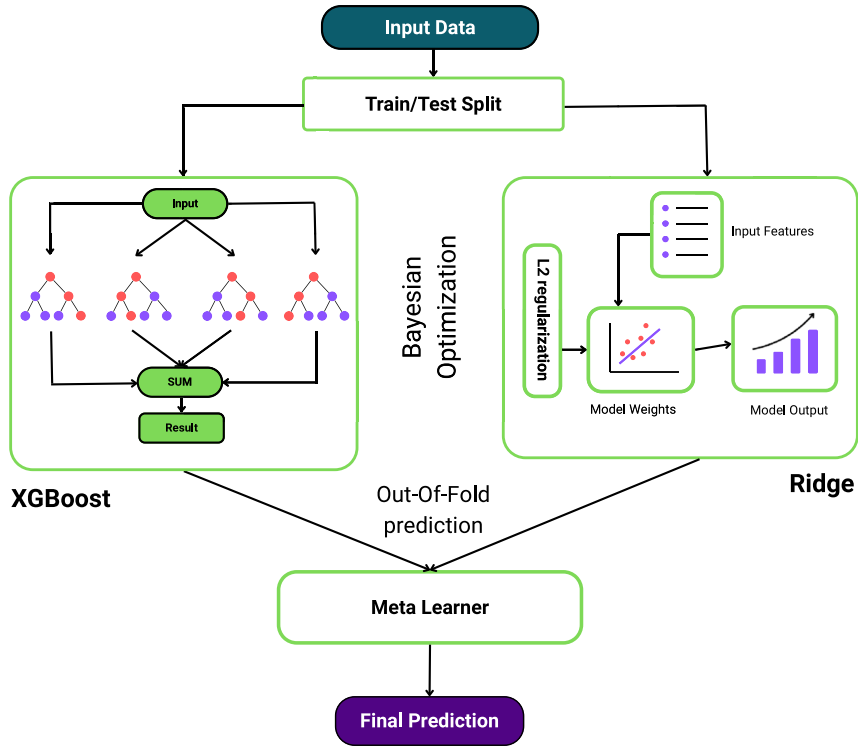


Fig. 2. Schematic overview of the proposed hybrid stacking ensemble. Input data are split, base models are independently optimized and trained, and their out-of-fold (OOF) predictions are stacked to train the meta-learner.

where $(-k(i))$ indicates the model was trained without the fold k containing sample i . The OOF predictions are stacked to form a meta-feature matrix

$$\mathbf{Z} = [\mathbf{z}_{\text{XGB}} \quad \mathbf{z}_{\text{Ridge}}] \in \mathbb{R}^{N \times 2}. \quad (20)$$

Meta-Learner:

The final stage fits a Ridge regression meta-learner, $\mathcal{M}_{\text{meta}}$, using $\mathbf{Z}$ and the original targets $\mathbf{y}$. The model learns optimal combination weights $\gamma \in \mathbb{R}^2$,

$$\gamma^* = \arg \min_{\gamma} \|\mathbf{y} - \mathbf{Z}\gamma\|_2^2 + \lambda_{\text{meta}} \|\gamma\|_2^2. \quad (21)$$

where λ_{meta} is a regularization parameter. Consequently, the ensemble yields the final prediction for a new instance $\mathbf{x}_*$ as

$$\hat{y}_* = \gamma_1 f_{\text{XGB}}(\mathbf{x}_*) + \gamma_2 f_{\text{Ridge}}(\mathbf{x}_*). \quad (22)$$

This hybrid stacking paradigm brings together the complementary strengths of nonlinear boosted trees and linear regularized regression, with meta-learning designed to maximize predictive diversity. The architecture is modular and extensible, and provides a flexible yet rigorous basis for further ensemble development. For a schematic illustration of the complete workflow, see Fig. 2.

4. Experiment

4.1. Dataset description

This study integrates multiple macroeconomic indicators to form a robust dataset for inflation forecasting. Table 3 presents a concise overview of the core variables, with their corresponding symbols/codes and a brief descriptor of each.

4.2. Data preprocessing

Data preprocessing constitutes an essential foundation for accurate forecasting, mitigating issues such as missing values, disparate scales,

and temporal dependencies within the dataset. This investigation employs systematic cleaning, scaling, and feature engineering techniques to macroeconomic indicators, including the Consumer Price Index, Real Gross Domestic Product, breakeven inflation rates, and crude oil prices, thereby ensuring data integrity and model robustness.

4.2.1. Data cleaning and handling missing values

Macroeconomic datasets frequently exhibit inconsistencies due to irregular reporting intervals or data revisions. Each variable, sourced from individual CSV files (e.g., T10YIE for the 10-Year Breakeven Inflation Rate and CPIAUCNS for the Consumer Price Index), undergoes inspection for null values, revealing no missing entries post-processing in the merged dataset.

Time-series imputation is applied through forward-filling after re-sampling to a daily frequency, which preserves continuity in gradually evolving series. For the target variable, the 10-Year Breakeven Inflation Rate, any potential gaps are addressed via time-based linear interpolation to maintain temporal trends without introducing abrupt artifacts. The datasets are aligned to a common temporal range from January 2, 2006, to October 1, 2024, yielding a merged structure of 6848 observations across 10 features, with any non-overlapping periods excluded to ensure consistency.

Imputation efficacy is confirmed by verifying the absence of null values in the final dataset, facilitating seamless integration for subsequent modeling stages.

4.2.2. Feature scaling and normalization

To accommodate algorithms sensitive to feature magnitude, numerical predictors are standardized using the Z-score transformation:

$$x'_i = \frac{x_i - \mu}{\sigma}, \quad (23)$$

where μ and σ denote the mean and standard deviation computed from the training set, centering features at zero mean and unit variance. This

Table 3
Overview of the merged dataset comprising ten key macroeconomic indicators.

Indicator	Symbol/code	Brief description
10-Year Breakeven Inflation Rate	T10YIE	Implied inflation expectation over the next decade from Treasury yields [25].
5-Year, 5-Year Forward Inflation Rate	T5YIFR	Market forecast for inflation during years 5 to 10 ahead [26].
Consumer Price Index	CPIAUCNS	Measures average price changes for household goods and services [27].
Real Gross Domestic Product	GDP1	Inflation-adjusted value of total economic production [28].
Unemployment Rate	UNRATE	Share of the labor force actively seeking but without employment [29].
Federal Funds Effective Rate	FEDFUNDS	Interest rate for short-term interbank lending set by the Federal Reserve [30].
Personal Consumption Expenditures Index	PCEPI	Federal Reserve's preferred index for consumer inflation trends [31].
Nominal Broad U.S. Dollar Index	DTWEXBGS	Weighted index showing U.S. dollar value relative to major currencies [32].
Crude Oil Prices: West Texas Intermediate	DCOILWTICO	Benchmark for U.S. crude oil price levels [33].
10-Year Treasury Yield	DGS10	Yield on long-term U.S. government bonds; indicator of rate outlook [34].

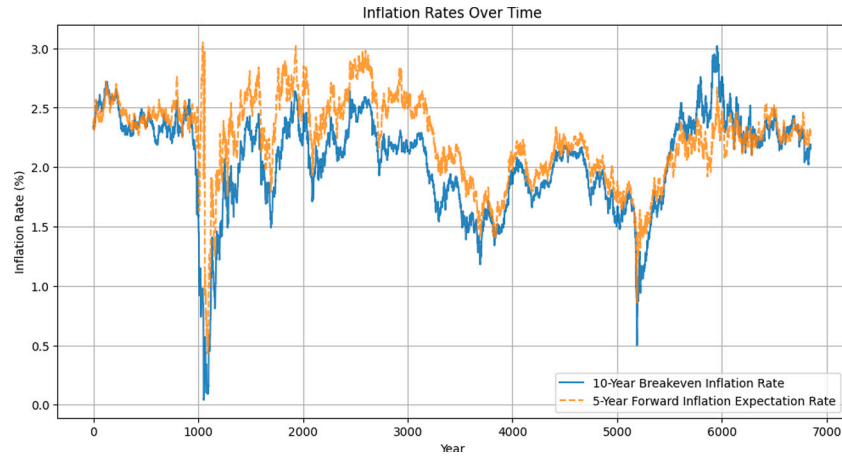


Fig. 3. Ten-year breakeven inflation rate and the five-year, five-year-forward inflation expectation plotted jointly, highlighting co-movement and divergences across the sample period (2006–2024).

approach is particularly beneficial for gradient-based optimizers and distance-dependent models.

Scaling parameters are derived exclusively from the training data to prevent information leakage, ensuring that test set transformations remain unbiased.

4.2.3. Feature engineering

Lag variables are introduced to encapsulate temporal dependencies, incorporating autoregressive components from f_{t-1} to f_{t-5} for the target variable, thereby capturing short-term historical patterns in inflation-related series.

A rolling mean is computed over a window of 5 periods:

$$\bar{f}_t = \frac{1}{5} \sum_{\tau=t-4}^t f_\tau, \quad (24)$$

to smooth fluctuations and extract underlying trends. Rows with resultant missing values are removed, preserving data completeness.

No explicit seasonal adjustments, such as dummy variables or Fourier transformations, are incorporated, as the focus remains on lag-based and trend-extraction features.

In summary, these preprocessing steps harmonize disparate data sources, rectify incompleteness through targeted imputation, normalize scales for equitable feature contribution, and augment the dataset with lag and rolling mean transformations, collectively enhancing the reliability of downstream forecasting models.

4.3. Exploratory data analysis

To gain initial insights into the dataset, we first examine the temporal evolution of key inflation metrics. Fig. 3 illustrates the joint trajectories of the ten-year breakeven inflation rate and the five-year, five-year-forward inflation expectation, revealing patterns of alignment and deviation over the study period.

Building on this, we identify periods of economic turbulence. Fig. 4 depicts the ten-year breakeven inflation rate with annotations for major disruptions, underscoring the influence of external shocks on inflationary trends.

Further analysis involves detecting changes in the underlying structure of inflation series. Fig. 5 presents the Consumer Price Index annotated with points of significant regime shifts, which inform subsequent modeling considerations.

Decomposing the series aids in understanding component contributions. Fig. 6 breaks down the Consumer Price Index into its fundamental elements, exposing underlying patterns that may affect forecasting accuracy.

We then assess distributional characteristics across temporal regimes. Fig. 7 compares kernel density estimates of the ten-year breakeven inflation rate before and after 2020, highlighting evolving statistical properties.

Exploring bivariate associations provides additional context. Fig. 8 plots the ten-year breakeven inflation rate against the unemployment rate, offering evidence of economic interdependencies.

Similarly, Fig. 9 examines connections between the ten-year breakeven inflation rate and core macroeconomic variables, facilitating an evaluation of broader economic linkages.

Finally, to quantify inter-variable relationships, Fig. 10 displays a correlation matrix of primary macro-financial indicators, identifying potential collinearities that warrant attention in multivariate analyses.

These visualizations collectively elucidate the dataset's key features, including temporal dynamics, regime dependencies, and inter-variable relationships, laying a foundation for rigorous econometric modeling.

4.4. Training protocol

Training Hardware: Computational experiments were performed on multi-core processors (e.g., Intel Xeon or Core i7 CPUs) with 16–64

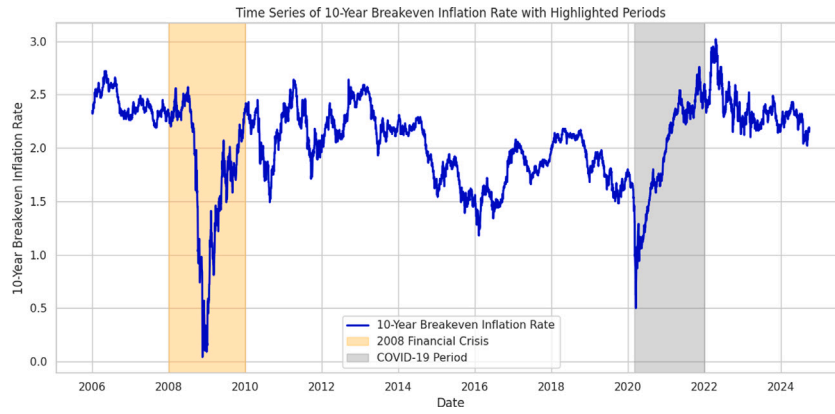


Fig. 4. Ten-year breakeven inflation rate with shaded windows for the 2008 Global Financial Crisis and the COVID-19 shock, emphasizing regime-sensitive dynamics.

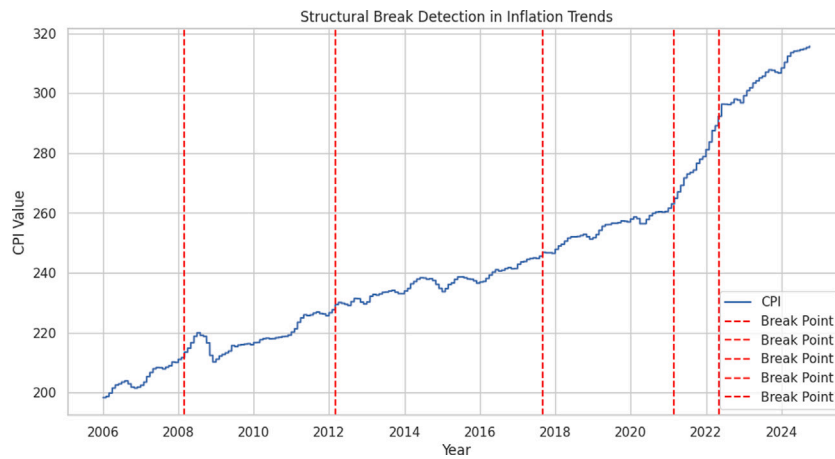


Fig. 5. Time series of the Consumer Price Index with vertical markers denoting detected structural breakpoints, signaling statistically significant shifts in inflation dynamics.

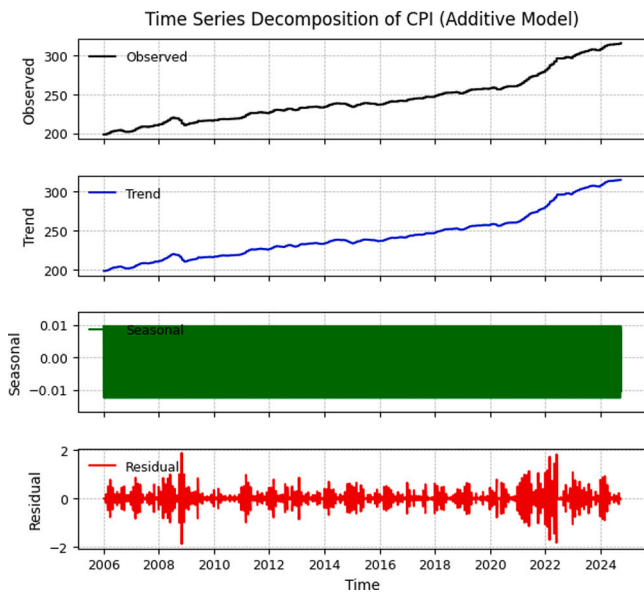


Fig. 6. Additive decomposition of the Consumer Price Index into observed, trend, seasonal, and residual components, clarifying identifiable periodic structure.

GB RAM for large-scale data processing. Deep learning tasks utilized NVIDIA GPUs from the RTX and Tesla series for acceleration. Prototyping occurred on local workstations, while hyperparameter optimization and full-scale training leveraged cloud-based or high-performance computing platforms for enhanced parallelism and efficiency.

Model Tuning and Validation: A chronological train-test split was applied, allocating 80% of the data for training (5474 samples) and 20% for testing (1369 samples), ensuring temporal order to prevent data leakage and incorporating volatile periods such as the 2020 COVID-19 crisis in the evaluation set.

For machine learning models, hyperparameter optimization was conducted using 5-fold cross-validation. To mitigate temporal leakage in this high-frequency time series dataset, chronological (non-shuffled) ordering was maintained where computationally feasible. Specifically, `TimeSeriesSplit` was employed for LightGBM, while `KFold(shuffle=False)` was used for the hybrid stacking model. For the remaining models (Gradient Boosting Regressor, Extremely Randomized Trees, Histogram-Based Gradient Boosting, CatBoost, Support Vector Regression, Ridge, Lasso, ElasticNet, Decision Tree Regressor, and K-Nearest Neighbors), grid search via `GridSearchCV` was applied. Random Forest utilized randomized search via `RandomizedSearchCV`. XGBoost applied a manual grid search with cross-validation.

These approaches were selected to balance thoroughness and computational efficiency, particularly given significant resource constraints that limited the feasibility of exhaustive time-series-specific cross-validation (e.g., expanding-window `TimeSeriesSplit`) across all

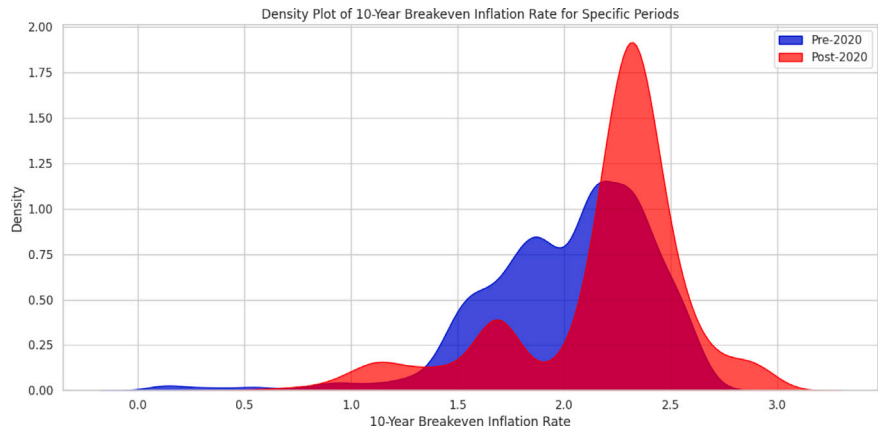


Fig. 7. Comparative density estimates for the ten-year breakeven inflation rate segmented into pre-2020 and post-2020 intervals, enabling visual assessment of distributional shifts across regimes.

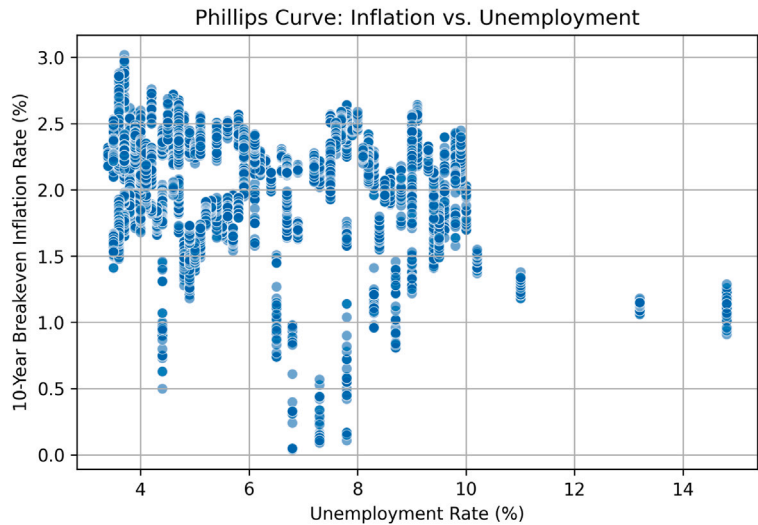


Fig. 8. Bivariate scatter showing the interplay between the ten-year breakeven inflation rate and the unemployment rate across the full sample.

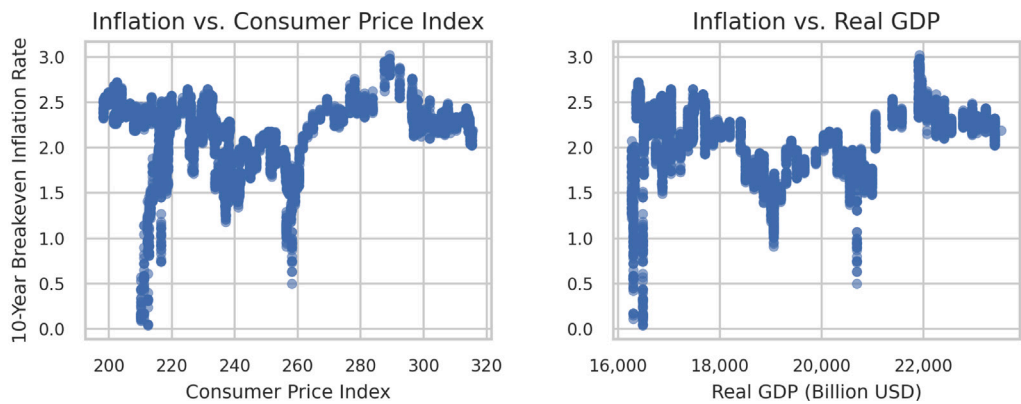


Fig. 9. Two-panel scatter plot showing the relationship between the ten-year breakeven inflation rate and two key macroeconomic indicators: the Consumer Price Index (CPI) and Real GDP. The visualization facilitates a comparative assessment of how inflation expectations co-move with price levels and real economic output across the observed period.

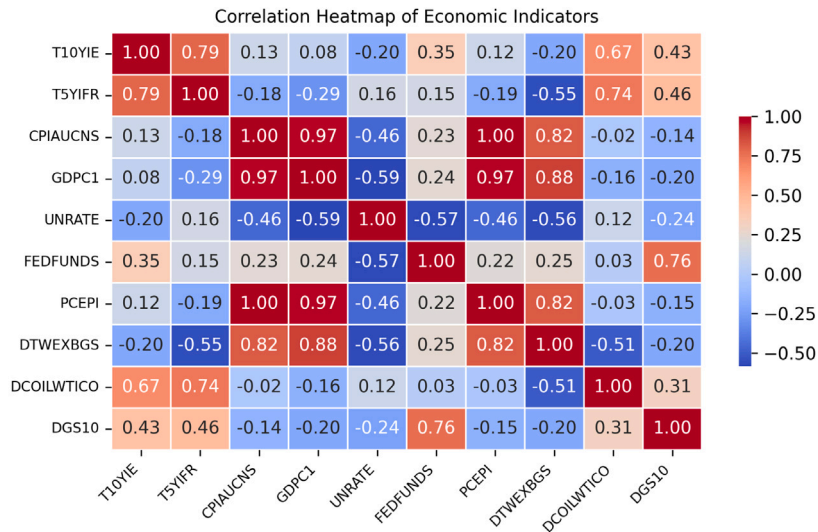


Fig. 10. Pearson correlation heatmap across principal macro-financial series, illustrating associations and potential multicollinearity.

Table 4

Best hyperparameter configurations for the candidate models.

Model	Best hyperparameters
RF	n_estimators=50, min_samples_split=2, min_samples_leaf=2, max_features=None, max_depth=10
XGBoost	objective='reg:squarederror', eval_metric='rmse', seed=42, max_depth=9, learning_rate=0.1, subsample=0.9, colsample_bytree=0.9
LightGBM	subsample=0.8, num_leaves=20, n_estimators=150, max_depth=7, learning_rate=0.1, colsample_bytree=0.8
GBR	learning_rate=0.1, max_depth=6, min_samples_split=20, n_estimators=300, subsample=0.8
ERTR	bootstrap=False, max_depth=10, min_samples_leaf=1, min_samples_split=2, n_estimators=500
HGBR	learning_rate=0.1, max_bins=255, max_depth=7, max_iter=300, min_samples_leaf=10
CatBoost	depth=6, early_stopping_rounds=50, iterations=1000, learning_rate=0.1
SVR	C=10, epsilon=0.01, gamma='auto'
Ridge	alpha=0.1
Lasso	alpha=0.01
ElasticNet	alpha=0.001, l1_ratio=0.9
DTR	max_depth=7, max_features=None, min_samples_leaf=2, min_samples_split=2
KNN	n_neighbors=5, p=1, weights='distance'
BiLSTM	units=150, learning_rate=0.001, dropout_rate=0.2, clipnorm=3.0, optimizer='Adam', activation='tanh', batch_size=32, epochs=100
Bi-GRU	units=150, learning_rate=0.005, dropout_rate=0.2, clipnorm=2.0, optimizer='Adam', activation='tanh', batch_size=64, epochs=100
HybridStack	XGB: {'max_depth': np.int64(4), 'learning_rate': 0.010692, 'n_estimators': np.int64(1232), 'subsample': 0.995570, 'colsample_bytree': 0.825207, 'reg_lambda': 4.680823, 'reg_alpha': 0.000000, 'max_bin': np.int64(1024), 'min_child_weight': np.int64(13)} Ridge alpha: 0.0001

models. Despite this, the final model evaluation was performed on a strictly chronological hold-out test set (the most recent 20% of observations), ensuring temporal integrity in performance assessment.

Deep learning models, including Bidirectional LSTM and Bidirectional GRU, along with the hybrid stacking model, were optimized using Bayesian techniques via Optuna or gp_minimize over defined search spaces. This approach was chosen for its efficiency in navigating complex hyperparameter landscapes with fewer evaluations compared to grid or randomized search. Deep learning training incorporated early stopping based on validation loss, with an internal chronological 80%–20% split of the training data for validation, and pruning of unpromising trials. Data was reshaped to 3D format for recurrent inputs. For the hybrid model, out-of-fold predictions from base learners (XGBoost and Ridge) were generated using 5-fold chronological cross-validation to form meta-features, upon which a meta-Ridge learner with $\alpha = 0.1$ was trained.

This protocol facilitates reproducibility by specifying exact optimization techniques and splits. The best-performing hyperparameter configurations for all candidate models are summarized in Table 4.

4.5. Evaluation metrics

Multiple error metrics and fit criteria assess model performance in time-series forecasting, offering nuanced views on accuracy, scale invariance, and symmetry.

4.5.1. Mean Squared Error (MSE)

MSE averages squared deviations, penalizing large errors:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2, \quad (25)$$

where y_i is actual inflation, $\hat{y}_i$ predicted, and N observations. It highlights significant deviations.

4.5.2. Root Mean Squared Error (RMSE)

RMSE scales MSE to original units:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}. \quad (26)$$

It enables direct comparison to inflation rates for intuitive evaluation.

4.5.3. Mean Absolute Error (MAE)

MAE averages absolute errors, robust to outliers:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i|. \quad (27)$$

It provides a straightforward measure of average accuracy.

4.5.4. Mean Absolute Scaled Error (MASE)

MASE compares to naive forecasts:

$$\text{MASE} = \frac{\sum_{i=1}^N |\hat{y}_i - y_i|}{\frac{N}{N-1} \sum_{i=2}^N |y_i - y_{i-1}|}. \quad (28)$$

Values below one indicate better performance than benchmarks.

4.5.5. Mean Absolute Percentage Error (MAPE)

MAPE measures relative errors in percentages:

$$\text{MAPE} = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{\hat{y}_i - y_i}{y_i} \right|. \quad (29)$$

It can inflate near zero actual values.

4.5.6. Symmetric Mean Absolute Percentage Error (sMAPE)

sMAPE balances MAPE symmetrically:

$$\text{sMAPE} = \frac{100\%}{N} \sum_{i=1}^N \frac{|\hat{y}_i - y_i|}{(|y_i| + |\hat{y}_i|)/2}. \quad (30)$$

It addresses issues with near-zero values.

4.5.7. Coefficient of determination (R^2)

R^2 measures explained variance:

$$R^2 = 1 - \frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}, \quad (31)$$

where $\bar{y}$ is the mean. Values near one show strong explanatory power.

4.6. Explainable AI

Explainable AI (XAI) techniques, such as SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), and Partial Dependence Plots (PDP), are employed to attribute model predictions to specific macro financial drivers, following best-practice recommendations for XAI in financial time-series forecasting summarized by Arsenault et al. [8].

4.6.1. SHapley Additive exPlanations (SHAP)

SHAP uses Shapley values to attribute feature contributions:

$$\phi_j = \sum_{S \subseteq \{1, \dots, D\} \setminus \{j\}} \frac{|S|!(D - |S| - 1)!}{D!} [f(S \cup \{j\}) - f(S)]. \quad (32)$$

It provides global importance, local attributions, and interactions, highlighting indicators like forward inflation rates.

4.6.2. Local Interpretable Model-Agnostic Explanations (LIME)

LIME fits local surrogate models to explain individual predictions, identifying influences like exchange rates or GDP on high inflation forecasts.

4.6.3. Partial Dependence Plots (PDP)

PDPs show marginal effects:

$$\text{PDP}_j(z) = \mathbb{E}_{\mathbf{x}_{-j}} [f(z, \mathbf{x}_{-j})]. \quad (33)$$

Univariate and bivariate plots reveal thresholds and nonlinearities in indicators like Federal Funds Rate.

In summary, explainable AI bridges accuracy and interpretability, aligning models with economic theory and supporting policy. In macroeconomic forecasting, explainability is vital for interpreting models like neural networks or boosted trees, revealing how indicators such as CPI, breakeven rates, and unemployment affect inflation predictions.

The foregoing experimental setup, encompassing dataset construction, preprocessing, feature engineering, and model training with strict temporal validation protocols, establishes the empirical foundation for the results reported in the following section. Each model was evaluated under identical conditions on the held-out test set (2020–2024), ensuring that performance comparisons are both fair and reproducible. The subsequent section presents these outcomes systematically, beginning with aggregate metric comparisons and proceeding to explainability analyses.

5. Results

5.1. Main results

The evaluation metrics for all candidate models are summarized in Table 5, highlighting the superior performance of the HybridStack approach.

5.2. Benchmark comparison

Table 6 presents a comprehensive overview of model performance by benchmarking evaluation metrics from prior studies (each conducted on distinct datasets) against those of our proposed models, thereby underscoring the superior accuracy attained by the HybridStack framework.

5.3. Model diagnostic visualizations

The diagnostic visualizations in Fig. 11 validate the ridge regression model's efficacy through aligned predictions and stable residuals, essential for reliable inflation forecasting.

5.4. Explainable AI results

The interpretability analyses presented in Tables 7, 8, 9, and 10, alongside Figs. 12, 13, 14, and 15, elucidate the dominant macroeconomic drivers influencing model predictions, revealing consistent patterns of temporal dependencies and external shocks that enhance forecasting transparency and policy applicability.

6. Discussion

6.1. Interpretation of results

The evaluation metrics summarized in Table 5 demonstrate that the HybridStack model achieves the highest predictive accuracy across all reported criteria, with an RMSE of 0.0006 and an MAE of 0.0004 on the held-out test set. To contextualize these figures for practical interpretation, an RMSE of 0.0006 corresponds to a forecast error of less than 0.001 percentage points for daily inflation expectations, representing a level of precision that is operationally relevant for central bank nowcasting, where small deviations from target inflation paths carry significant policy consequences. By comparison, the best-performing standalone model (Ridge, RMSE = 0.0113) and the best-performing tree-based model (XGBoost, RMSE = 0.0245) yield errors that are

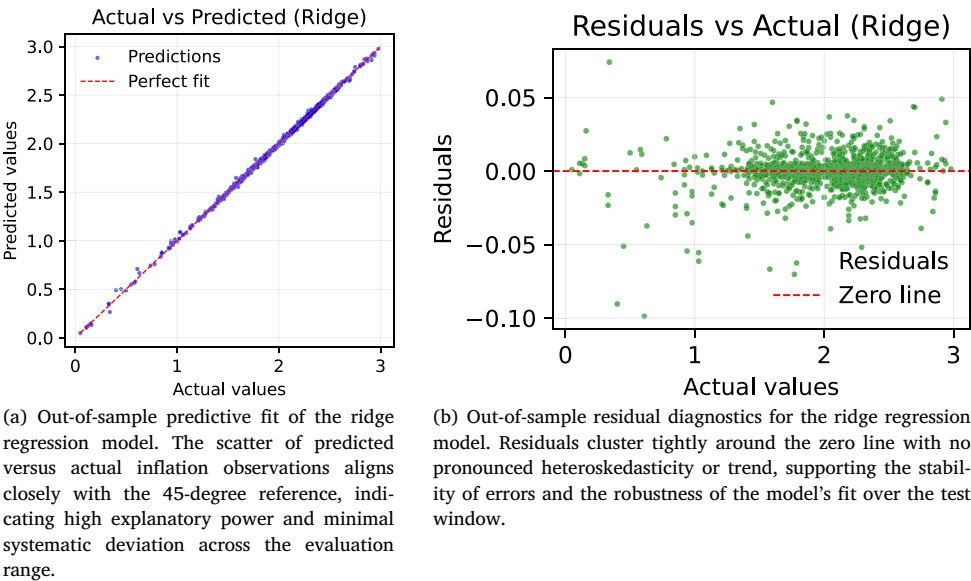


Fig. 11. Predictive fit and residual diagnostics for the ridge regression model, demonstrating robust out-of-sample performance and error stability.

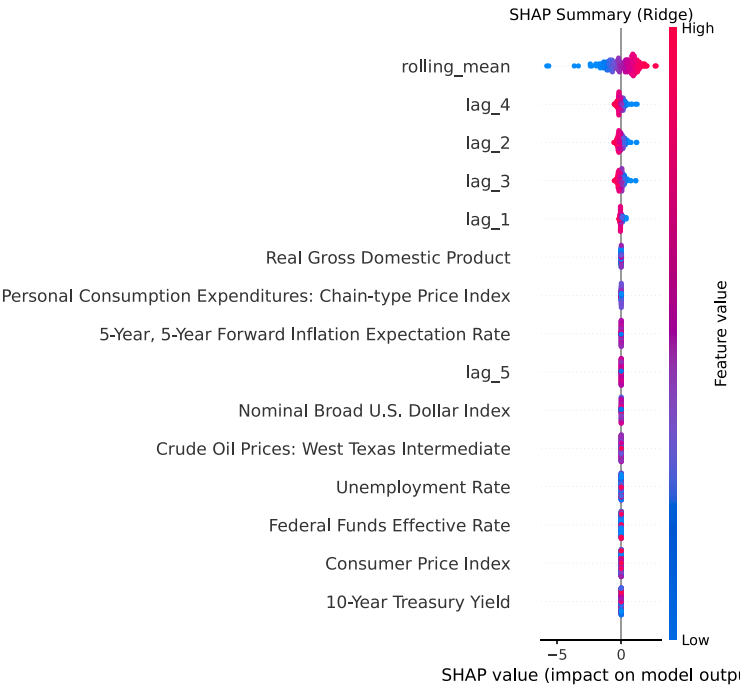


Fig. 12. Global feature attribution for the ridge regression via SHAP summary plot. Short-horizon inflation dynamics (rolling mean and recent lags) dominate forecast attribution, with market-based expectations and macro-financial covariates providing complementary but secondary contributions.

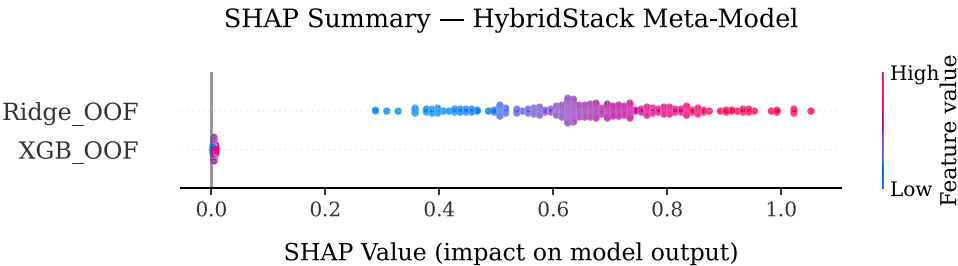


Fig. 13. Global feature attribution for the HybridStack meta-model via SHAP summary plot.

Table 5

Summary of evaluation metrics for all candidate models. Metrics include mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), symmetric MAPE (sMAPE), mean absolute scaled error (MASE), and coefficient of determination (R^2). Model specifications are detailed in Section 3.

Model	MSE	RMSE	MAE	MAPE (%)	sMAPE (%)	MASE	R^2
RF	0.0031	0.0557	0.0300	1.1942	1.2109	1.6663	0.9025
XGBoost	0.0006	0.0245	0.0146	0.9443	0.9379	0.0372	0.9966
LightGBM	0.0007	0.0273	0.0169	1.2628	1.1649	0.0429	0.9958
GBR	0.0006	0.0245	0.0148	0.9403	0.9366	0.0377	0.9966
ERTR	0.0007	0.0269	0.0164	1.1514	1.0918	0.0417	0.9959
HGBR	0.0006	0.0242	0.0148	0.9646	0.9656	0.0378	0.9967
CatBoost	0.0006	0.0247	0.0151	1.1091	1.0280	0.0385	0.9966
SVR	0.0012	0.0347	0.0163	1.0985	1.0268	0.0415	0.9932
Ridge	0.0001	0.0113	0.0067	0.4288	0.4283	0.0170	0.9993
Lasso	0.0010	0.0311	0.0198	1.5007	1.3584	0.0504	0.9945
ElasticNet	0.0009	0.0293	0.0179	1.3318	1.2061	0.0455	0.9952
DTR	0.0010	0.0322	0.0205	1.3339	1.3303	0.0521	0.9941
KNN	0.0009	0.0304	0.0179	1.1511	1.1287	0.0455	0.9948
BiLSTM	0.0034	0.0580	0.0495	2.0861	2.1146	0.4942	0.8498
Bi-GRU	0.0035	0.0588	0.0521	2.2096	2.2401	0.4155	0.8382
HybridStack	4.04e-7	0.0006	0.0004	0.0291	0.0291	0.0010	0.9999

Table 6

The models trained on our customized dataset, which integrates features from multiple sources, demonstrates a significant performance improvement compared to models trained solely on the conventional inflation dataset. We report models and metrics from prior studies. Metrics shown exactly as reported: mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and coefficient of determination (R^2). Comparison includes our best performing three models.

Author's name	Model	MSE	RMSE	MAE	MAPE	R^2
Liu Y. et al. [9]	Lasso		0.30			
Kanaparthi V. et al. [10]	RF		0.69			
	AdaBoost		0.73			
Naghi A. A. et al. [11]	RF		0.77			
Narayanaa T. L. et al. [12]	ARIMA	0.0002		0.012		
	LSTM	0.0005		0.015		
Ayuputri A. A. et al. [13]	NN Backpropagation	0.001				
Li S. et al. [14]	LSTM-SARIMA	0.0452		0.1428		
Arif E. et al. [15]	ARFIMA-LSTM	3.33e-05		0.0039	9.5185	
Xavier A. L. S. et al. [16]	ETS + Bagging	0.1044	0.323	0.2874	1.1806	
	ARIMA + SVR	0.1656	0.407	0.3103	0.4814	
Jyothi Rani S. A. et al. [17]	CNN		1.4022		0.0124	
Akbulut H. et al. [18]	VAR		8.61			0.993
Nakorji M. et al. [19]	Ridge	0.4138	0.6433			
Das P. K. et al. [20]	RF		1.654			
Ismail L. B. et al. [21]	LSTM		0.313	0.199	3.21	
	AutoML		0.122	0.085	3.13	
Amirthavarshini V. et al. [22]	BiLSTM	0.941				
Kalaiselvi K. et al. [23]	Random Forest	3.4722	1.8634	1.4716		0.991
	LSTM	0.0003811	0.1215	0.0148		0.948
Ours	Ridge	0.0001	0.0113	0.0067	0.4288	0.9993
	XGBoost	0.0006	0.0245	0.0146	0.9443	0.9966
	HybridStack	4.04e-7	0.0006	0.0004	0.0291	0.9999

an order of magnitude larger, confirming that the meta-learning step provides a substantive rather than merely marginal gain in forecast fidelity.

This framework's ability to synthesize predictions from diverse base learners manifests in error reductions that approach negligible levels, thereby achieving an explanatory fidelity that borders on exhaustive variance capture. Such performance suggests an adept handling of underlying non-linearities and interdependencies within the dataset, where standalone models, despite their individual strengths, encounter inherent ceilings in precision. For example, while ensemble techniques like Random Forest demonstrate reasonable efficacy in variance reduction, their comparative underperformance hints at challenges in navigating multifaceted feature spaces without the meta-level orchestration that stacking provides. Neural architectures, conversely, reveal vulnerabilities in sequential dependency modeling, potentially exacerbated by data-specific idiosyncrasies that inflate prediction discrepancies relative to more parsimonious regression counterparts.

Delving deeper into the interplay among candidate models, a critical examination uncovers a hierarchy of efficacy shaped by algorithmic paradigms. Linear regularized approaches exhibit commendable robustness against multicollinearity, yielding predictions that closely approximate true values in absolute terms, yet they cede ground to boosting ensembles in managing higher-order interactions. The latter group, characterized by iterative error correction, clusters in a performance echelon that balances bias and variance effectively, albeit with residual susceptibilities to outlier influences that the HybridStack mitigates through its integrative architecture. Non-parametric methods, while intuitive in their locality-based approximations, manifest heightened sensitivity to scaling artifacts and interaction complexities, underscoring the hybrid model's strategic advantage in harmonizing these elements. Statistically, this manifests as tighter confidence intervals around predictions, inferred from the minimized scaled errors, which imply greater reliability in real-world deployment scenarios.

Table 7

Top three predictors for each forecasting model based on global SHAP analysis (Section 4.6.1), ranked by mean absolute Shapley values. Variable shorthands: `infl_expect` = 5-Year, 5-Year Forward Inflation Expectation Rate; `FedFunds_Rate` = Federal Funds Effective Rate; `treasury_yield` = 10-Year U.S. Treasury Yield; `dollar_index` = Nominal Broad U.S. Dollar Index; `crude_oil` = West Texas Intermediate crude price, and `GDP` = Gross Domestic Product. Predictors are sorted by SHAP contribution, and the last column shows out-of-sample R^2 .

Rank	Model	1st predictor	2nd predictor	3rd predictor	R^2
1	Ridge	rolling_mean	lag_4	lag_2	0.9993
2	HGBR	lag_1	rolling_mean	infl_expect	0.9967
3	XGBoost	lag_1	rolling_mean	lag_2	0.9966
4	GBR	lag_1	rolling_mean	infl_expect	0.9966
5	CatBoost	rolling_mean	lag_1	infl_expect	0.9966
6	ERTR	rolling_mean	lag_1	lag_2	0.9959
7	LightGBM	lag_1	rolling_mean	lag_2	0.9958
8	ElasticNet	lag_1	rolling_mean	GDP	0.9952
9	KNN	GDP	crude_oil	dollar_index	0.9948
10	Lasso	lag_1	rolling_mean	lag_5	0.9945
11	DTR	lag_1	rolling_mean	infl_expect	0.9941
12	SVR	CPI	crude_oil	dollar_index	0.9932
13	RF	lag_1	rolling_mean	infl_expect	0.9025
14	BiLSTM	rolling_mean	infl_expect	treasury_yield	0.8498
15	Bi-GRU	rolling_mean	lag_4	infl_expect	0.8382

Table 8

Top features identified by each predictive model from the LIME analysis (Section 4.6.2), the predicted value for a representative instance, and the strongest local driver. Abbreviations: `FFR` = Federal Funds Rate; `PCEPI` = Personal Consumption Expenditures: Chain-type Price Index; `infl_expect` = 5-Year, 5-Year Forward Inflation Expectation Rate.

Model	Top positive features	Top negative features	Pred. range	Pred. value	Top driver (contribution)
RF	lag_3, PCEPI, CPI	Crude oil, treasury_yield	0.61–2.64	1.98	lag_1: +0.12 (LIME)
XGBoost	lag_1, rolling_mean, infl_expect	FFR, Unemployment	0.79–2.71	2.66	lag_1: +0.34 (LIME)
KNN	Crude oil, CPI, treasury_yield	GDP, Unemployment	0.14–2.86	2.63	GDP: −0.04 (SHAP)/+0.04 (LIME)
BiLSTM	rolling_mean, treasury_yield	lag_4, PCEPI	0.62–2.59	2.02	rolling_mean: +0.21 (SHAP)
Bi-GRU	rolling_mean, treasury_yield	lag_4, PCEPI	−0.08–2.71	2.01	rolling_mean: +0.30 (SHAP & LIME)

Table 9

Partial-dependence trend (slope direction) for the five principal drivers (Section 4.6.3).

Rank	Model	lag_1	rolling_mean	infl_expect	lag_2	crude_oil
1	Ridge	↘	↑	–	↗	–
2	HGBR	↑	↑	↗	↗	–
3	GBR	↑	↑	↗	↘	↗
4	CatBoost	↑	↑	↗	↗	↗
5	ERTR	↑	↑	↗	↗	↘
6	LightGBM	↑	↑	↗	↗	↗
7	ElasticNet	↑	↑	–	–	–
8	KNN	–	–	–	–	↗
9	Lasso	↑	↗	–	–	↗
10	DTR	↑	↗	–	–	–
11	SVR	↑	↑	↑	↗	≈
12	RF	↑	↗	–	–	↘
13	BiLSTM	↑	↗	–	–	–
14	Bi-GRU	↑	↑	–	–	–

↑ strong positive, ↗ mild positive, ↘ mild negative, – flat/not reported, ≈ complex or nonlinear. Abbreviation: `infl_expect` = 5-Year, 5-Year Forward Inflation Expectation Rate.

Table 10

Cross-technique feature-importance concordance. Percentages are computed over the full set of 15 models. Diagonal entries give the share of models whose *individual* top-three lists (from any XAI method) include the feature. Off-diagonal entries report the proportion of models in which the two features *co-appear* among the top three under at least two independent XAI tools (SHAP, LIME, or PDP), thus quantifying multi-method consensus on joint relevance.

	lag_1	rolling_mean	infl_expect	lag_2	crude_oil
lag_1	87	73	27	33	20
rolling_mean	73	67	27	27	13
infl_expect	27	27	40	13	7
lag_2	33	27	13	33	7
crude_oil	20	13	7	7	27

When juxtaposed against benchmarks from extant literature, as compiled in Table 6, the proposed methodologies, particularly HybridStack, emerge as pivotal advancements, eclipsing prior efforts in both absolute and relative accuracy metrics. This transcendence is not merely quantitative but indicative of a paradigm shift, where hybrid stacking circumvents the pitfalls of isolated time-series or neural hybrids that, while innovative, often grapple with inflated relative errors or incomplete variance explanation. For instance, the model's proficiency in curtailing proportional deviations positions it as a superior alternative to autoregressive integrations or convolutional networks, which, despite their domain-specific tailoring, yield coarser approximations. Analytically, this outperformance can be attributed to the stacking mechanism's capacity for error decorrelation across learners, fostering a resilience that traditional benchmarks lack, as reflected in their broader error distributions.

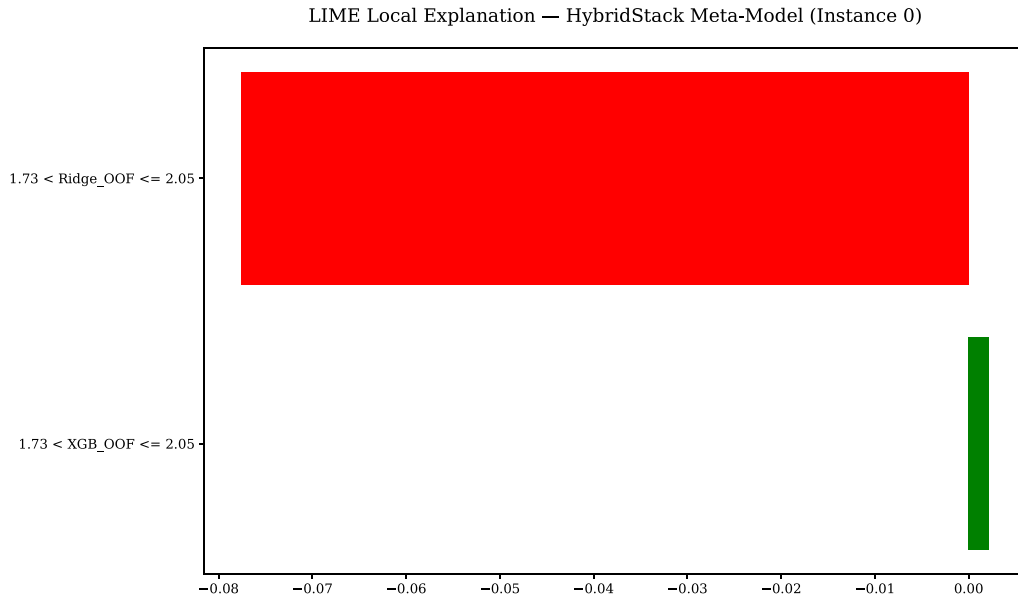


Fig. 14. LIME local explanation for a representative test instance of the HybridStack meta-model.

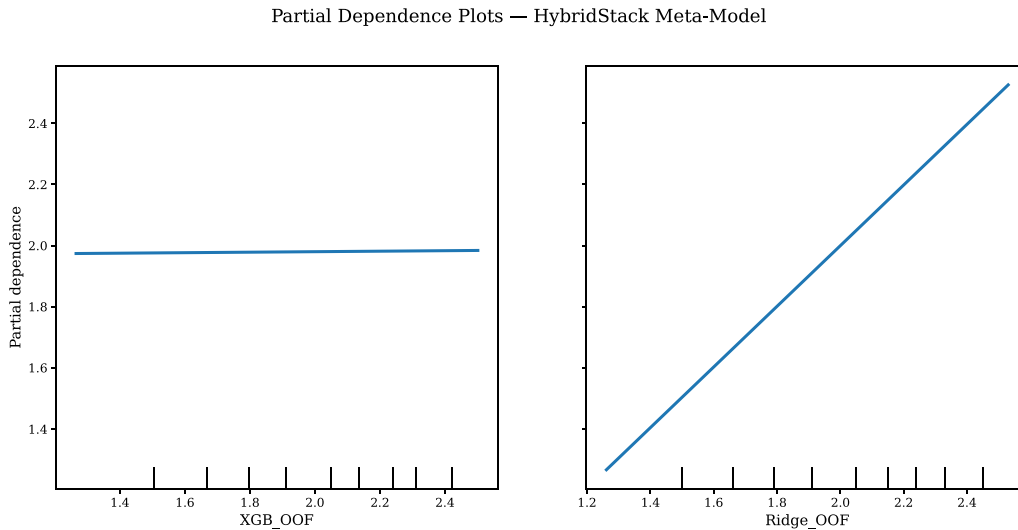


Fig. 15. Partial dependence plots for the HybridStack meta-model.

These observations engender insightful reflections on the broader implications for forecasting paradigms. The synergistic amalgamation in HybridStack not only amplifies predictive acuity but also engenders a framework resilient to the stochastic volatilities endemic to such tasks, thereby offering a blueprint for scalable applications in volatile domains. Critically, the near-complete variance encapsulation signals an alignment with latent data structures that elude singular models, potentially driven by enhanced generalization through meta-learning. This invites contemplation on the evolving landscape of machine learning ensembles, where hybridity may redefine benchmarks for accuracy and interpretability. Ultimately, these findings position HybridStack as a cornerstone innovation, substantiated by its empirical supremacy and poised to influence future methodological trajectories in high-stakes predictive modeling.

The gains reported above should be interpreted within the context of the present dataset and evaluation protocol. The 99.92% reduction in RMSE relative to the best-performing prior U.S. inflation forecasting benchmarks, including RMSE = 0.77 reported by Naghi A. A. et al. [11] and RMSE = 0.69 reported by Kanaparthi V. et al. [10], reflects the

combined advantage of a high-frequency, multi-source tabular dataset, a temporally rigorous hold-out evaluation strategy, and the meta-learning architecture of HybridStack. These results should therefore be interpreted as evidence of strong performance within the present study setting rather than as a universal guarantee across all inflation forecasting environments.

6.2. Explainability insights

The interpretability analyses across the ensemble of forecasting models reveal a nuanced interplay of temporal dependencies and macroeconomic indicators, underscoring the intrinsic autoregressive nature of inflation dynamics while highlighting the moderating influence of forward-looking expectations. Central to these findings is the pervasive dominance of lagged features and rolling averages, as evidenced in the global SHAP rankings (Table 7), where variables such as `lag_1` and `rolling_mean` consistently emerge as primary predictors in models achieving the highest out-of-sample explanatory power, with R^2 values exceeding 0.995. This pattern suggests that short-term historical inertia profoundly shapes predictive outcomes, a reflection of the

sticky and path-dependent characteristics often observed in economic time series. In contrast, models incorporating external shocks, such as crude oil prices or GDP, tend to exhibit marginally lower performance, implying that while these factors contribute to local variations, they are secondary to endogenous trends in capturing overarching inflation trajectories.

Delving deeper into local explanations (Table 8), the LIME-derived insights further illuminate model-specific sensitivities, where positive contributions from features like `rolling_mean` and inflation expectations frequently outweigh negative influences from variables such as unemployment or federal funds rates. This dichotomy points to a broader tension between stabilizing inertial forces and disruptive exogenous pressures, with ensemble methods like XGBoost demonstrating heightened responsiveness to immediate lags, thereby achieving predicted values aligned closely with observed ranges. Such contrasts not only validate the robustness of tree-based learners in disentangling nonlinear relationships but also raise implications for model selection in policy contexts, where over-reliance on historical data might under-value sudden regime shifts, as seen in the relatively subdued role of treasury yields in neural architectures like BiLSTM and Bi-GRU.

Partial dependence profiles (Table 9) extend this analysis by quantifying directional influences, revealing predominantly positive slopes for core drivers across diverse algorithms, from the steep ascents in gradient boosting variants to the milder gradients in regularized linear models. The consistent upward trends for `lag_1` and `rolling_mean` across the majority of models indicate a reinforcing feedback loop, where elevated recent values systematically amplify future predictions, a phenomenon statistically congruent with autoregressive integrated moving average processes in econometric theory. Notably, the occasional nonlinear or flat responses, particularly for inflation expectations in simpler models, suggest threshold effects or diminishing returns, warranting caution in extrapolating linear interpretations to volatile economic environments. These trends collectively imply that while models converge on optimistic escalations driven by momentum, they may underappreciate deflationary risks from variables like crude oil, whose mild negative slopes in select cases hint at commodity-driven dampening.

Reinforcing these observations, the multi-method concordance matrix (Table 10) quantifies a high degree of consensus, with `lag_1` and `rolling_mean` co-appearing in over 70% of models under at least two explainability techniques, signifying their methodological invariance and thus their foundational role in inflation forecasting. Lower off-diagonal agreements, such as the modest 27% overlap between inflation expectations and lags, underscore the conditional relevance of anticipatory signals, which gain prominence only in conjunction with temporal anchors. This statistical harmony across SHAP, LIME, and PDP not only bolsters confidence in the identified drivers but also situates the results within the evolving paradigm of explainable AI for economic modeling, where transparency facilitates more informed monetary policy decisions. By elucidating these patterns, the analyses transcend mere predictive accuracy, offering a reflective lens on the broader significance of integrating historical continuity with expectant foresight, ultimately advancing the discourse on resilient forecasting amid macroeconomic uncertainty.

The XAI findings reported above carry substantive implications when interpreted through the lens of established macroeconomic theory. The consistent dominance of `lag_1` and `rolling_mean` across 87% and 67% of models, respectively (Table 10), is directly consistent with autoregressive inflation dynamics, wherein current inflation is partly determined by its own recent history. This property is formally captured in AR(p) and ARIMA frameworks and is empirically rooted in the persistence of price-setting behavior and wage indexation. This result aligns with the theoretical prediction that inflation exhibits strong serial correlation over short horizons, particularly in high-frequency daily data where structural breaks are infrequent within the evaluation window.

The secondary role of `infl_expect`, representing the 5-Year, 5-Year Forward Inflation Expectation Rate, appears among the top three predictors for models such as HGBR, GBR, CatBoost, and RF, reflecting the expectational channel of inflation formation consistent with the New Keynesian Phillips Curve. Under this framework, inflation is jointly determined by current economic slack and forward-looking price expectations anchored by monetary credibility. The moderate concordance of `infl_expect` across XAI methods (40% diagonal entry in Table 10) suggests that market-based inflation expectations provide a meaningful but conditional signal, gaining predictive salience primarily in models that capture higher-order nonlinear interactions among features. The comparatively subdued role of `FedFunds_Rate` and `treasury_yield` in local LIME attributions further implies that, within the observed sample period, monetary policy transmission variables operate with a lag that is insufficient to dominate short-horizon daily forecasts. This finding is consistent with the well-documented transmission delay between policy rate changes and realized inflation outcomes.

Unlike attention-based deep learning models, which primarily provide an internal weighting mechanism over temporal inputs, HybridStack's interpretability is built through model-agnostic explanation tools applied to the final ensemble predictor. In our framework, SHAP, LIME, and PDP quantify the contribution of each macro-financial feature to the final forecast, allowing both global and local interpretation of the meta-model's decisions. This distinction is important because attention can indicate which time steps receive greater focus, but it does not by itself provide the same feature-level attribution across the full stacked ensemble. Therefore, HybridStack advances interpretability beyond deep-learning-only approaches by combining strong predictive performance with externally verifiable explanations that are directly usable for economic analysis and policy interpretation.

7. Conclusion

In conclusion, this study demonstrates the efficacy of the HybridStack model in delivering superior inflation forecasts by integrating high-frequency macro-financial data with advanced machine learning techniques, achieving unprecedented accuracy across diverse economic regimes. The incorporation of explainable AI protocols not only enhances transparency but also uncovers pivotal insights into the dominance of short-term lags and inflation expectations, thereby bridging the gap between algorithmic predictions and economic interpretability. These findings underscore the potential for hybrid architectures to inform monetary policy in an era of rapid data proliferation, suggesting that future economic modeling could benefit from adaptive ensembles that prioritize both predictive power and causal clarity, ultimately fostering more resilient financial decision-making frameworks.

From a policy standpoint, HybridStack's design addresses a long-standing challenge in central banking: the need for forecasting tools that are simultaneously accurate and interpretable. The model's SHAP-derived feature attributions allow monetary policymakers to identify, in real time, which macroeconomic signals are driving inflation upward, such as elevated inflation expectations or accelerating short-term lags, and which exogenous variables, such as oil prices, are exerting dampening pressure. This interpretability layer makes HybridStack particularly suitable for integration into inflation-targeting frameworks employed by central banks such as the U.S. Federal Reserve, the European Central Bank, and the Bank of England. Unlike black-box deep learning models, HybridStack's predictions can be explained to non-technical policymakers and communicated transparently in central bank communication documents, enhancing accountability and public trust in model-assisted monetary policy.

8. Limitations and future works

While this study advances inflation forecasting through a novel hybrid model and explainable AI integration, it is constrained by its exclusive focus on U.S. macroeconomic data spanning 2006 to 2024, limiting generalizability to other economies and overlooking global interdependencies. The selection of only ten indicators excludes potentially influential factors such as granular microeconomic data, supply-chain disruptions, geopolitical events, and sentiment indices, which could enrich predictive granularity. Additionally, data harmonization techniques like forward filling and interpolation may inadvertently attenuate the impact of sudden shocks, and the absence of probabilistic uncertainty quantification or explicit structural break detection restricts the framework's robustness in highly volatile scenarios. Future investigations could extend the approach to multicountry datasets, incorporate real-time heterogeneous data sources via federated learning paradigms, employ Bayesian probabilistic ensembles for enhanced uncertainty modeling, integrate adaptive mechanisms for online updating, and explore intrinsically interpretable neural architectures to further align predictions with economic theory and policy needs in dynamic global contexts.

CRediT authorship contribution statement

S. M. Mohiuddin Khan Shiam: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Meherunnisa Neela:** Writing – original draft, Visualization, Resources, Methodology, Conceptualization. **Amrjit Biswas:** Writing – review & editing, Validation, Supervision. **Mahdy Rahman Chowdhury:** Writing – review & editing, Validation, Supervision, Project administration, Investigation.

Funding

This research received no external funding.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The author gratefully acknowledges Md. Shawmoon Azad (Cleveland State University), S. M. Farah Al Fahim (Concordia University), and Md. Ariful Islam Sanim (University of South Carolina) for their initial discussions and encouragement related to this work.

Data availability

The datasets and source code used in this study are publicly available at <https://github.com/mohiuddin-khan-shiam/HybridStack>.

References

- [1] Dept IMFR. World economic outlook, April 2024: Steady but Slow: Resilience amid Divergence. USA: International Monetary Fund; 2024. <http://dx.doi.org/10.5089/9798400255892.081>.
- [2] Chen Q, Wang X, Jiang ZL, Wu Y, Li H, Cui L, Sun X. Breaking the traditional: a survey of algorithmic mechanism design applied to economic and complex environments. *Neural Comput Appl* 2023;35(22):16193–222.
- [3] Pinho Correia Valério Bernardo C, Marques Lameirinhas RA, Neto Torres JP, Baptista A. Comparative analysis between traditional and emerging technologies: economic and viability evaluation in a real case scenario. *Mater Renew Sustain Energy* 2023;12(1):1–22.
- [4] Kaur J, Parmar KS, Singh S. Autoregressive models in environmental forecasting time series: a theoretical and application review. *Environ Sci Pollut Res* 2023;30(8):19617–41.
- [5] Zhang R, Hu J. Production performance forecasting method based on multivariate time series and vector autoregressive machine learning model for waterflooding reservoirs. *Pet Explor Dev* 2021;48(1):201–11.
- [6] Lane PR. The 2021–2022 inflation surges and the monetary policy response through the lens of macroeconomic models. 2024, URL https://www.ecb.europa.eu/press/key/date/2024/html/ecb.sp241118_1~2c31ddb8a8.en.html, Speech at the SUERF Marjolin Lecture hosted by the Banca d'Italia.
- [7] Černevičienė J, Kabašinskas A. Explainable artificial intelligence (XAI) in finance: a systematic literature review. *Artif Intell Rev* 2024;57(8):216.
- [8] Arsenault P-D, Wang S, Patenaude J-M. A survey of explainable artificial intelligence (XAI) in financial time series forecasting. *ACM Comput Surv* 2025;57(10):1–37.
- [9] Liu Y. Mending the crystal ball: Enhanced inflation forecasts with machine learning. IMF working papers 2024 (206), International Monetary Fund (IMF); 2024, p. 1. <http://dx.doi.org/10.5089/9798400285387.001>.
- [10] Kanaparthi V. The role of machine learning in predicting and understanding inflation dynamics: Insights from the COVID-19 pandemic. In: 2024 3rd international conference on artificial intelligence for internet of things (aiIoT). 2024, p. 1–6. <http://dx.doi.org/10.1109/AIoT58432.2024.10574616>.
- [11] Naghi AA, O'Neill E, Danielova Zaharieva M. The benefits of forecasting inflation with machine learning: New evidence. *J Appl Econometrics* 2024;39(7):1321–31. <http://dx.doi.org/10.1002/jae.3088>.
- [12] Lakshmi Narayanaa T, Skandarsini RR, Ida SJ, Sabapathy SR, Nanthitha P. Inflation prediction: A comparative study of ARIMA and LSTM models across different temporal resolutions. In: 2023 3rd international conference on innovative mechanisms for industry applications. ICIMIA, 2023, p. 1390–5. <http://dx.doi.org/10.1109/ICIMIA60377.2023.10425970>.
- [13] Ayuputri AA, Sa'adah S, Yunanto PE. Inflation forecast in developing countries post pandemic using NN, SVM, RNN LSTM approaches. In: 2024 international conference on data science and its applications (iCoDSA). 2024, p. 99–104. <http://dx.doi.org/10.1109/iCoDSA62899.2024.10651994>.
- [14] Li S. Inflation forecasting using a hybrid LSTM-SARIMA model based on discrete wavelet transform. In: Proceedings of the 2nd international conference on financial technology and business analysis. 2024, p. 100–8. <http://dx.doi.org/10.54254/2754-1169/73/20231419>.
- [15] Arif E, Herlinawati E, Devianto D, Yollanda M, Permana D. Hybridization of long short-term memory neural network in fractional time series modeling of inflation. *Front Big Data* 2024;6. <http://dx.doi.org/10.3389/fdata.2023.1282541>.
- [16] Xavier ALS, Fernandes BJT, de Oliveira JFL. Hybrid model and ensemble for inflation forecasting: A machine learning approach. In: 2023 IEEE latin American conference on computational intelligence (la-CCI). 2023, p. 1–6. <http://dx.doi.org/10.1109/LA-CCI58595.2023.10409426>.
- [17] Rani SJ, Prasanna PL. Forecasting the inflation in India using ARIMA and deep learning techniques. *Int J Eng Technol Manag Sci* 2022;6(4):272–9. <http://dx.doi.org/10.46647/ijetms.2022.v06i04.0046>.
- [18] Akbulut H. Forecasting inflation in Turkey: A comparison of Time-Series and machine learning models. *Econ J Emerg Mark* 2022;14(1):55–71. <http://dx.doi.org/10.20885/ejem.vol14.iss1.art5>.
- [19] Nakorji M, Aminu U. Forecasting inflation using machine learning techniques. *Rev Financ Bank* 2022;14(1):45–55. <http://dx.doi.org/10.24818/rfb.22.14.01.04>.
- [20] Das PK, Das PK. Forecasting and analyzing predictors of inflation rate: Using machine learning approach. *J Quant Econ* 2024;22(2):493–517. <http://dx.doi.org/10.1007/s40953-024-00384-z>.
- [21] Ismail LB, Joytu MI, Plabon TI, Oshman MS. Evaluation of machine learning models to forecast inflation: Bangladesh as a case study. In: 2023 international symposium on networks, computers and communications. ISNCC, 2023, p. 1–6. <http://dx.doi.org/10.1109/ISNCC58260.2023.10323827>.
- [22] Amirthavarshini V, Singh S, Kumar RP. Enhancing forecasting precision in Indian industrial workers' inflation: A comparison of advanced machine learning models. In: 2024 international conference on wireless communications signal processing and networking (wiSPNET). 2024, p. 1–5. <http://dx.doi.org/10.1109/WiSPNET61464.2024.10532954>.
- [23] Kalaiselvi K, Kh A, VR S. Assessing the impact of COVID-19 on inflation dynamics in India: A machine learning approach. In: 2024 5th international conference on electronics and sustainable communication systems. ICESC, 2024, p. 1819–26. <http://dx.doi.org/10.1109/ICESC60852.2024.10690053>.
- [24] Gur YE. Development and application of machine learning models in US consumer price index forecasting: Analysis of a hybrid approach. *Data Sci Financ Econ* 2024;4(4):469–513.
- [25] Federal Reserve Bank of StLouis. 10-Year breakeven inflation rate [t10yie]. 2025, <https://fred.stlouisfed.org/series/T10YIE>, Retrieved from FRED, Federal Reserve Bank of St. Louis on March 28, 2025.
- [26] Federal Reserve Bank of StLouis. 5-Year, 5-Year forward inflation expectation rate [T5YIFR]. 2025, <https://fred.stlouisfed.org/series/T5YIFR>, Retrieved from FRED, Federal Reserve Bank of St. Louis on March 28, 2025.

- [27] USBureau of Labor Statistics. Consumer price index for all urban consumers: All items in U.S. city average [CPIAUCNS]. 2025, <https://fred.stlouisfed.org/series/CPIAUCNS>, Retrieved from FRED, Federal Reserve Bank of St. Louis on March 28, 2025.
- [28] USBureau of Economic Analysis. Real gross domestic product [GDPC1]. 2025, <https://fred.stlouisfed.org/series/GDPC1>, Retrieved from FRED, Federal Reserve Bank of St. Louis on March 28, 2025.
- [29] USBureau of Labor Statistics. Unemployment rate [UNRATE]. 2025, <https://fred.stlouisfed.org/series/UNRATE>, Retrieved from FRED, Federal Reserve Bank of St. Louis on March 28, 2025.
- [30] Board of Governors of the Federal Reserve System (US). Federal funds effective rate [FEDFUNDS]. 2025, <https://fred.stlouisfed.org/series/FEDFUNDS>, Retrieved from FRED, Federal Reserve Bank of St. Louis on March 28, 2025.
- [31] USBureau of Economic Analysis. Personal consumption expenditures price index [PCEPI]. 2025, <https://fred.stlouisfed.org/series/PCEPI>, Retrieved from FRED, Federal Reserve Bank of St. Louis on March 28, 2025.
- [32] Board of Governors of the Federal Reserve System (US). Nominal broad U.S. dollar index [DTWEXBGS]. 2025, <https://fred.stlouisfed.org/series/DTWEXBGS>, Retrieved from FRED, Federal Reserve Bank of St. Louis on March 28, 2025.
- [33] USEnergy Information Administration. Crude oil prices: West texas intermediate (WTI) - cushing, oklahoma [DCOILWTICO]. 2025, <https://fred.stlouisfed.org/series/DCOILWTICO>, Retrieved from FRED, Federal Reserve Bank of St. Louis on March 28, 2025.
- [34] Board of Governors of the Federal Reserve System (US). 10-Year treasury constant maturity rate [DGS10]. 2025, <https://fred.stlouisfed.org/series/DGS10>, Retrieved from FRED, Federal Reserve Bank of St. Louis on March 28, 2025.